

What Does It Mean to Be Black? Experimental Evidence on the Situational Deployment of Racial Schemas

Sociology of Race and Ethnicity

1–21

© American Sociological Association 2026

DOI: 10.1177/23326492261469562

journals.sagepub.com/home/sre



Marissa E. Thompson¹, A. J. Alvero²,
Sam Trejo³, and Daphne O. Martschenko⁴

Abstract

How do Black Americans construct narratives about the meaning and boundaries of Blackness in America today? We leverage an original, nationally representative survey experiment among U.S.-born Black Americans ($n = 3,100$) to examine (1) justifications used to validate or reject another's racial self-identification as Black and (2) how situational cues shape these narratives. Using a vignette experiment and structural topic modeling, we find that both the social context (e.g., a census form vs. scholarship application) and the prior self-identification of the vignette individual affect the topics invoked by respondents, while genetic ancestry test information has no meaningful treatment effects. All told, we demonstrate that racial schemas—including frameworks emphasizing race as a personal choice or invoking the one-drop rule—are situationally deployed rather than fixed. We also highlight the limits of standard quantitative measures of racial classification, which do not entirely capture the nuanced ways that underlying narratives about race and belonging develop in contemporary Americans' minds.

Keywords

Black Americans, racial identity, genetic ancestry tests, experimental methods, racial classification

Introduction

Americans hold a range of complex views about race: what defines group membership, who has the authority to classify others, and how to evaluate individuals who do not cleanly fall into a single racial category (Abascal et al. 2025; Roth 2016; Schachter, Flores, and Maghbouleh 2021; Thompson et al. 2026).¹ These beliefs shed light on the breadth and complexity of racial schemas, the cognitive structures underlying the ways in which people make meaning of race and how they construct rules for applying racial

categories to themselves and others (Roth 2012). From a theoretical perspective, racial schemas highlight the ways that ordinary people participate in the social construction of race and the upholding of racial hierarchies.

¹Columbia University, New York, NY, USA

²Cornell University, Ithaca, NY, USA

³Princeton University, NJ, USA

⁴Stanford University, CA, USA

Corresponding Author:

Marissa E. Thompson, Department of Sociology,
Columbia University, New York, NY 10027-6902, USA.
Email: marissa.thompson@columbia.edu

Narratives around what it means to be a member of a particular racial group and who can—or cannot—self-identify in a particular way are inherently cultural in nature, illustrating on a broad scale how people interpret their own racial position, how they organize status and power along racial lines, and which aspects of similarity or difference they choose to emphasize as they evaluate others (Morning 2011; Omi and Winant 2014; Roth 2012, 2016; Warikoo and Novais 2015). Racial schemas, then, constitute the micro-level processes through which racial boundaries are maintained and contested on a macro level, forming the basis for shared understandings about the nature of race in today's society.

And yet, understanding the wide range of beliefs about the nature of race is a complex task (Roth 2016). Much of the quantitative literature on this topic tends to operationalize and measure racial processes in a relatively coarse way, such as by using Likert scales measuring agreement with certain beliefs or discrete responses to racial classification prompts. While these approaches provide important insight into key aspects of social norms, they are often not able to capture the multifaceted and dynamic nature of racial processes. Ethnographies and interview studies tap into this nuance, but they tend to capture the views of a specific subset of a population rather than a nationally representative pool. As a result, while we now know much about the cues and characteristics that shape how Americans, on average, classify others, there is still much to learn about the process through which these judgments are made.

Are racial schemas fixed or situationally deployed? In this study, we examine the range of racial schemas that Black Americans hold about the meaning of being Black in contemporary America. We focus specifically on Black Americans due to their unique place in the U.S. racial structure: historically subjected to hypodescent laws that defined any presumed African ancestry as determinative, yet at present often navigating tensions between phenotype, ancestry, and cultural authenticity in an era where genetic ancestry testing is widespread. Both historically and in the

present day, the racial boundary defining Blackness in the United States is distinctive—stemming, in part, from the legal enshrinement of “one-drop” rules (Davis 2001), whereby any presumed African ancestry resulted in classification as Black (regardless of one's self-identification). While there are no legal restrictions on self-identifications in the U.S. today, studies have illustrated a range of characteristics that influence beliefs about the nature of Blackness, including results from genetic ancestry tests (GATs; Abascal et al. 2025; Ghoshal 2025; A. Nelson 2016; Schachter et al. 2021; Thompson et al. 2026).

Drawing from literatures on racial boundaries and the schemas that uphold them, we investigate how a range of contexts and cues shape how Black Americans conceptualize the meaning of race and justify their classifications of others. To do so, we leverage an original, nationally representative survey experiment of U.S.-born Black American adults ($n = 3,100$; see Thompson et al. 2026). Our study reveals how historically marginalized groups themselves actively negotiate belonging and adaptively deploy particular norms in response to situational cues. More specifically, we use vignettes about a fictitious individual who has recently taken a GAT and now self-identifies as Black, while randomly varying three aspects of this individual's background: the level of sub-Saharan African (SSA) ancestry on their GAT result, whether they are identifying as Black on a census form or a scholarship application, and whether or not the fictitious individual self-identified as Black prior to taking the GAT. We then use a computational text analysis and topic modeling approach based on 6,200 free text responses across our 3,100 respondents (Grimmer, Roberts, and Stewart 2022; L. K. Nelson 2020; M. E. Roberts et al. 2014).

We estimate the causal effect of treatment assignment on the prevalence of various topics used by respondents to justify their opinions and classifications via free text responses. By moving beyond conventional racial classification and approval measures, our study adds nuance to sociological understandings of how people evaluate claims to Blackness and

craft their own definitions of what it means to be Black in America today. Further, our methodological approach allows us to capture the views of a larger and more representative sample than would typically be available when using traditional interview methods. Computational text analysis is by no means a replacement for traditional survey- or interview-based approaches, but it enables data collection and analysis at a scale that offers researchers the ability to identify subtleties that may be left uncovered by other approaches. At the same time, the rapid introduction of generative artificial intelligence, large language models (LLMs), and their associated chatbots introduces new concerns about the authenticity of text data collected from online survey participants (Zhang, Xu, and Alvero 2025). Fortunately, our study was fielded just as the first demo version of such tools was introduced to the public; as a result, our data reflects one of the last opportunities to collect a large corpus of text data without concerns about the new and unpredictable effect of respondent use of LLMs.

Across the free text responses, we identify the themes used to discuss and debate racial boundaries and belonging, with the most common topics relying on language that focuses on race as a personal choice and multiracial populations, among others. Still, we observe wide heterogeneity across responses and find a range of conceptual strategies used to validate or reject the vignette individual's racial identification. Our experiment reveals several surprising empirical patterns. First, we find no effect of GAT result information on how respondents discussed the vignette individual's decision—an unexpected result given a series of prior studies that have found large and meaningful effects of GAT information on classification outcomes (Schachter et al. 2021; Thompson et al. 2026). On the other hand, we find effects of both context and prior identification on the topics invoked by respondents as they grapple with their perceptions of the vignette individual and the social appropriateness of their self-identification. For example, we find that respondents in the census condition were more likely to emphasize

personal choice than those in the scholarship condition. Moreover, those reading about an individual who changed his or her racial identification to identify as Black were less likely to focus on physical features and “one-drop” rules than those who read about an individual with a stable self-identification (e.g., someone who has not changed their racial self-identification). This suggests that, not only do classifications shift meaningfully in response to vignette characteristics, but so too do respondents' justifications regarding the appropriateness of specific self-identification decisions.

Our results build upon prior work on the complex ways that race is understood and refracted within communities. Through the use of text analysis, we can add important nuance to existing studies of how boundaries are constructed and policed to illustrate how Black Americans flexibly deploy justifications in response to varying situational cues. Further, we follow calls from Schachter and Weisshaar (2025) and M. E. Roberts et al. (2014) to integrate computational text-as-data methods into experimental designs. In doing so, we highlight some of the limits of standard quantitative measures of agreement or classification, which do not always index the nuanced ways that classification logics take shape in the minds of contemporary Americans. Ultimately, our results suggest that racial schemas operate as flexible repertoires rather than fixed understandings of race, with individuals strategically deploying different justifications—such as personal choice, genetic ancestry, or even cultural experience—depending on whether the situation entails institutional stakes or boundary crossing.

Background

Race is a social construct that is based, in part, on a range of personal characteristics as well as the context in which a person is self-identifying and being evaluated by others (Morning 2011; Omi and Winant 2014; Roth 2016). While a large body of literature has examined the attributes that shape self-identification and subjective feelings of group membership, self-identifications do not unilaterally map on

to external classifications or acceptance by external observers. In this study, we focus on how three characteristics—GAT results, context, and prior identification—shape narratives around the perceived acceptability of another’s self-identification.

Some 60 million Americans (about one in five) have used at-home, direct-to-consumer kits to trace the approximate geographic regions where their ancestors likely lived (Orth 2022). Though the information provided in a GAT result is conceptually distinct from socially constructed notions of racial categories, the two are often conflated by consumers, who do not typically receive information about how to interpret their results. GATs are but one avenue through which social norms around race have shifted in recent decades, but they represent a theoretically generative case for how people make sense of seemingly “objective” information that may or may not align with their previous self-conceptions or conceptions of others (A. Nelson 2016; Roth and Ivermark 2018). Several studies have illustrated that Americans are sensitive to GAT information when classifying others (Schachter et al. 2021; Thompson et al. 2026). This is true both among Black Americans and among other groups, though Black Americans tend to have higher likelihoods of classifying racially ambiguous individuals as Black relative to White Americans (Thompson et al. 2026). While the effects of skin tone tend to outweigh the effects of GAT results and ancestry information in head-to-head experimental comparisons (Abascal et al. 2025; Schachter et al. 2021), increased amounts of SSA ancestry tend to increase the likelihood of external classification as Black. Further, even individuals with very small amounts of SSA ancestry on the GATs—on the order of 4 percent—are still classified as Black by Black observers about half of the time (Thompson et al. 2026).

We also examine the role of context in shaping the narratives deployed by Black Americans. Though prior work has illustrated that people are not sensitive to context when classifying others, it is not clear the extent to which justifications and narratives are shifted.

For example, Abascal et al. (2024) used a conjoint experiment to illustrate that context (self-identification on an anonymous survey compared to a scholarship application) did not meaningfully shape classifications by external observers. Likewise, Thompson et al. (2026) found that context (self-identification on a census form compared to a scholarship application) did not shape classification as Black, but it did affect respondents’ likelihoods of reporting that they approved of the self-identification decision of the vignette individual.

Finally, we examine whether crossing a boundary and changing one’s racial self-identification is viewed differently from those with more stable racial identifications (e.g., vignettes where the individual already identified as Black in the past compared to scenarios where they did not). Prior work has shown that self-identifications matter: people are more likely to classify someone as Black when they have self-identified as Black (Abascal et al. 2025). However, this is not always the case, as some characteristics also increase the likelihood that a particular self-identification will be rejected. Importantly, racial boundaries are heterogeneous: some are “bright,” others are “blurred,” and several have shifted in recent decades as political tides shift, official census categories are changed, and social movements emerge that emphasize new and different elements of belonging (Alba 2005; Bailey, Fialho, and Loveman 2018; Flores, Loria, and Casas 2023; Loveman and Muniz 2007; Thompson 2026). In this work, we specifically examine the boundary defining who can and cannot identify as Black, as judged socially by other Black observers. We focus on this particular racial boundary given the long legacy of the one-drop rule in shaping the Black-White binary, which information about GATs may cue. Further, given pressure to address racial inequalities in access to higher education, the boundary around Blackness is important for understanding how context—such as differences between a census and a potential affirmative action context such as a scholarship application—affects how people respond and characterize their reactions.²

Racial Classification and Schemas

From a theoretical perspective, this study examines in detail the racial schemas held by Black Americans as they judge who should be classified as Black. We draw from Roth (2012) in conceptualizing racial schemas as cognitive frameworks that bundle racial categories with understandings of their meanings, rules for application, and hierarchical ordering. Racial schemas translate abstract racial categories into practical judgments, making visible the boundary work underlying racial conceptions and sustaining racial stratification. The cognitive structures that underlie how people deploy and narrate classification rules are questions of culture and meaning-making (Pepin and Scarborough 2025; Valentino 2021; Valentino and Warren 2024). They reflect understandings about the nature of race that are perceived to be shared by other racial group members (Boutyline and Soter 2021; Brubaker 2016; Schachter et al. 2021; Thompson 2026), shedding light on how people make distinctions between in- and out-groups (Lamont and Molnár 2002). In drawing and maintaining racial/ethnic boundaries, groups on either side of a given divide negotiate understandings of membership and belonging through social interaction (Lamont and Molnár 2002; Omi and Winant 2014; Wimmer 2008).

Prior work has illustrated that people hold a range of beliefs about the nature and meaning of race and racial hierarchies (Morning 2009; Roth, van Stee, and Regla-Vargas 2023). Given the rigidity of the history of the one-drop rule in the United States, it is perhaps no surprise that some view race in essentialist terms: believing that race is an inherent attribute with a biological basis. Evidence from both the social and biological sciences has shown that there is scant evidence of a biological basis for race (Feldman and Lewontin 1975; Morning 2011; D. Roberts 2011; Roth et al. 2023; Yudell et al. 2016). Those with constructivist views of race define it as a social construct, characterizing both the meaning of race and the categories that divide racial groups as artifacts of a human process (Roth et al. 2023). While this view generally

reflects the one highlighted by social scientists, lay people do not always share this perception. Still, though essentialism and constructivism are often positioned as two opposing ends of a spectrum, the racial schemas held by the public are complex. Even beliefs understood to be essentialist—such as attributing race to direct genetic inheritance—can be strategically deployed rather than applied consistently across cases (Condit 2019; Haslam, Rothschild, and Ernst 2000).

However, prior work on how external observers classify others tends to focus exclusively on their classifications. While this is no doubt an important outcome, it leaves much to be desired about the nuance surrounding *how* people make subjective judgments about racial group membership. Whereas prior studies have shown the dimensions that affect classifications and other quantitative metrics associated with one's approval or disapproval of a vignette individual, what is less clear is *why* respondents react the way that they do. What are the contents of the narratives and justifications that Black Americans deploy when they evaluate whether someone can self-identify a particular way? Importantly, the qualitative, deliberative way that people justify their choices does not necessarily align with quick-response intuition captured by other methods (Lizardo et al. 2016; Vaisey 2009). Indeed, a long line of literature in sociology and cognitive science highlights that people's intuitions and instinctive responses about their values, culture, and motivations may—or may not—map onto the deliberative reasonings and the justifications that they offer to explain themselves (Jerolmack and Khan 2014; Miles, Charron-Chénier, and Schleifer 2019; Vaisey 2009). In other words, the way that people classify others and respond to close-ended questions about race may not align with the post-hoc reasonings that they use to explain their behavior to others (and even to themselves).

To illustrate this point, in this study, we compare the dimensions that shape racial classifications with those that shape written justifications. We follow Lizardo et al. (2016) in

Table 1. Summary Statistics of Survey Respondents.

	<i>M</i>	<i>SD</i>
Age	45.12	16.71
Female	0.53	
<i>Education</i>		
No high school	0.07	
High school graduate	0.39	
Some college	0.23	
Two-year degree	0.11	
Four-year degree	0.13	
Post-grad	0.08	
Household income (\$1,000)	47.30	58.64
<i>Immigration background</i>		
Parents & grandparents born in US	0.88	
Parents born in US	0.05	
Parents immigrants to US	0.07	
<i>Political Party</i>		
Democrat	0.62	
Republican	0.08	
Independent	0.21	
Other/not sure	0.10	
Observations	3,100	

Note. Weighted averages are displayed. Survey weights were provided directly by YouGov to approximate a nationally representative sample of U.S.-born Black American adults. Weights were created using a sampling frame generated from ACS using age, gender, race, and education, and then matched back to the respondent sample. Weights correspond to the inverse probability of selection into the sample.

conceptualizing these pathways as part of a broader domain of cultural thinking, which separates rapid observations (here, racial classifications) from more deliberate, slow, and rule-based reasonings (here, the written justifications provided via text data) to elaborate on how people process cues about another's race. In drawing on these literatures, we aim not to definitively map the specific cognitive processes that underlie racial classifications, but rather to illustrate that the dimensions that shape classifications need not map cleanly onto the dimensions that shape other forms of evaluation.

Methods

Experimental Design

To answer the questions posed by this study, we fielded a pre-registered survey experiment with a nationally representative sample of U.S.-born self-identified Black American adults in January

2023.³ Data for this study were contracted through YouGov. Descriptive statistics of the survey respondents (weighted to reflect a nationally representative sample of U.S.-born Black American adults) are displayed in Table 1. All demographic variables were collected directly by YouGov during the panel recruitment process. Portions of analyses drawn from this experiment have been published in separate articles (Thompson et al. 2026).

In each vignette, respondents were asked to evaluate the racial identification of a fictitious individual, where we randomly varied three characteristics of the vignette: (1) the individual's GAT results (4 percent vs. 36 percent SSA ancestry), (2) their prior racial self-identification (Black vs. non-Black), and (3) the context in which they identified as Black (census vs. scholarship). Respondents began the study by viewing an image of a fictitious individual (see Supplemental Appendix for images) and reading the following prompt:

Meet Aaron.⁴ Aaron is an American citizen whose family has been in the U.S. for generations. Aaron is usually unsure of how to identify, but mostly [has/ has not] identified as Black/African-American.

Aaron recently received his genetic ancestry test results from a company such as 23andMe or Ancestry.com. Genetic ancestry tests use a person's DNA to make an informed guess regarding where a person's ancestors lived. Aaron's genetic ancestry test results estimate that he has approximately [4% / 36%] Sub-Saharan African ancestry (from countries such as Ghana and Nigeria) and [96% / 64%] non-African ancestry.

A few weeks after receiving these genetic ancestry test results, Aaron [filled out a government census and identified as Black/African-American/applied for a college scholarship intended for Black/African-American students (e.g., NAACP)].

Selecting Images and Names. All of the images used in our experiment are drawn from the Chicago Face Database (Ma, Correll, and Wittenbrink 2015; Ma, Kantner, and Wittenbrink 2021). To select the six images presented alongside the vignette, we first used the Chicago Face Database norming data, which includes survey responses of perceived race and ethnicity for each image, to select a range of racially ambiguous images. We selected a preliminary set of images where the survey data displayed a reasonable level of perceived White-Black ambiguity. However, because our study focuses only on the perceptions of Black respondents, we pretested these images to ensure that Black Americans perceived them to be racially ambiguous (and that the images themselves were similar in their degree of White-Black ambiguity). As with the images, we selected racially ambiguous names (e.g., names that would not send a strong White or Black racial signal to respondents). We selected 12 total names (6 suggesting a woman and 6 suggesting a man) that were randomly assigned to vignettes and images.⁵ All results include both image and name fixed effects, which ensure that our treatment effects

are valid across a number of physical appearances and names. This broadens the generalizability of our work, as selecting only one image and name to use for the vignettes might lead our results to be specific to the particularities of the image and name selected.

In addition to the text outcomes described below, we also investigated responses to both a Likert-scale question on approval/disapproval of the vignette individual and racial classification (using a census-style list of racial/ethnic identifications to choose from). First, we measured how respondents reacted to the individual in the assigned vignette condition identifying as Black/African American in the specified context (measured using a 7-point Likert scale from “extremely negative” to “extremely positive”). Second, we asked respondents to provide a classification of the single best race/ethnicity (drawn from 2020 census categories) that they believed described the vignette individual. We investigate these outcomes in detail in a separate companion article (Thompson et al. 2026). For the purposes of comparison to analyses with the text outcomes, Table A1 in the Supplemental Appendix reproduces the main effects for from the nationally representative sample of 3,100 Black Americans for both classification and approval outcome (see Thompson et al. (2026) for additional details on quantitative outcomes). In addition, while the present study examines how situational cues shape narratives, in separate forthcoming work, we explore the moral dimensions of authenticity claims using a subset of the text data and a qualitative approach (Thompson et al., forthcoming).

Structural Topic Modeling

We solicited free responses to analyze using structural topic modeling (STM) text-as-data methods (Grimmer et al. 2022). Sociologists of culture and race/ethnicity, among others, have used computational tools such as STM to uncover schemas from large corpora of text data (Boutyline and Soter 2021; DiMaggio, Nag, and Blei 2013). However, sociologists

have been relatively slow to adopt the pairing of experimental methods and text-as-data approaches. In our case, this approach allows us to move beyond whether respondents approve or classify the vignette individual as Black, revealing how they justify these judgments (in addition to the underlying narratives they deploy in service of this boundary work). To solicit the free-response data used in this study, we included the following two prompts (which we refer to as the “Define Being Black” prompt and the “Reaction to Decision” prompt, respectively):

1. We are interested in your beliefs on what it means for a person to be Black/African-American. In your opinion, what are the characteristics, experiences, attitudes, and values that define being Black/African-American?
2. How do you feel about Aaron identifying as Black/African-American {on the government census / for a college scholarship intended for Black/African-American students} after taking a genetic ancestry test? Please specify your reactions, thoughts, and emotions about his decision.

Respondents provided written text in response to both questions. Table A2 in the Supplemental Appendix illustrates word distributions for both prompts.

STM, like other topic modeling methods, models word co-occurrence patterns across all of the documents in a given dataset (in our case, the free responses) to generate high-quality numerical representations of thematic content in text. Put simply, STM identifies clusters of words that tend to appear together across responses, which we interpret as underlying narrative themes or “topics.” This allows us to estimate the causal effect of the various vignette treatment conditions on the content of text responses. In doing so, we examine not only how individuals respond to classification and Likert-scale outcomes, but also how

different sets of information influence different types of free text responses.

Figure 1 displays our analytic approach to the text analysis. Following Egami et al. (2022), we use an *ex ante* train-test split approach to prevent overfitting and avoid the tautological issue of using training data to measure causal effects that arises when analyzing latent variables. We first randomly selected 50 percent of the data to use as our training set and the remaining 50 percent of our data for the testing set; this split was stratified to ensure equal representation of all of our various treatment groups in the training data. Importantly, while all respondents were given the option to provide text data, a very small number provided blank responses (seven responses across the two questions, from a total of 6,200 documents).

We preregistered our approach and STM model (which was estimated using only the training data, prior to accessing the test data). The text was pre-processed following standard approaches: all text was lowercased, stemmed (reduced to their root form such that words like “write,” “writing,” and “written” would all be reduced to “writ-”), and stopwords (high-frequency function words like “the” and “an”) were removed. We attended to the idiosyncrasies of the responses as well. For example, some respondents included emojis in their written responses (presumably written on their mobile phones), which were removed. We also simplified variations of some terms that appeared frequently into one word, such as transforming “African American” and “African-American” to “AfricanAmerican.” Percentage symbols were converted to “PERCENTAGE” in order to preserve their meaning in the topic modeling, and all references to the names of the vignette individuals (i.e., Aaron, Mia, Daniel) were collapsed to “EXPERIMENTNAME.”

After preprocessing the text, we next determined a suitable number of topics to generate for our final model. To determine the appropriate number of topics for our model, we used the *ldatuning* package in R (Murzintcev 2016). The *ldatuning* library generates topic

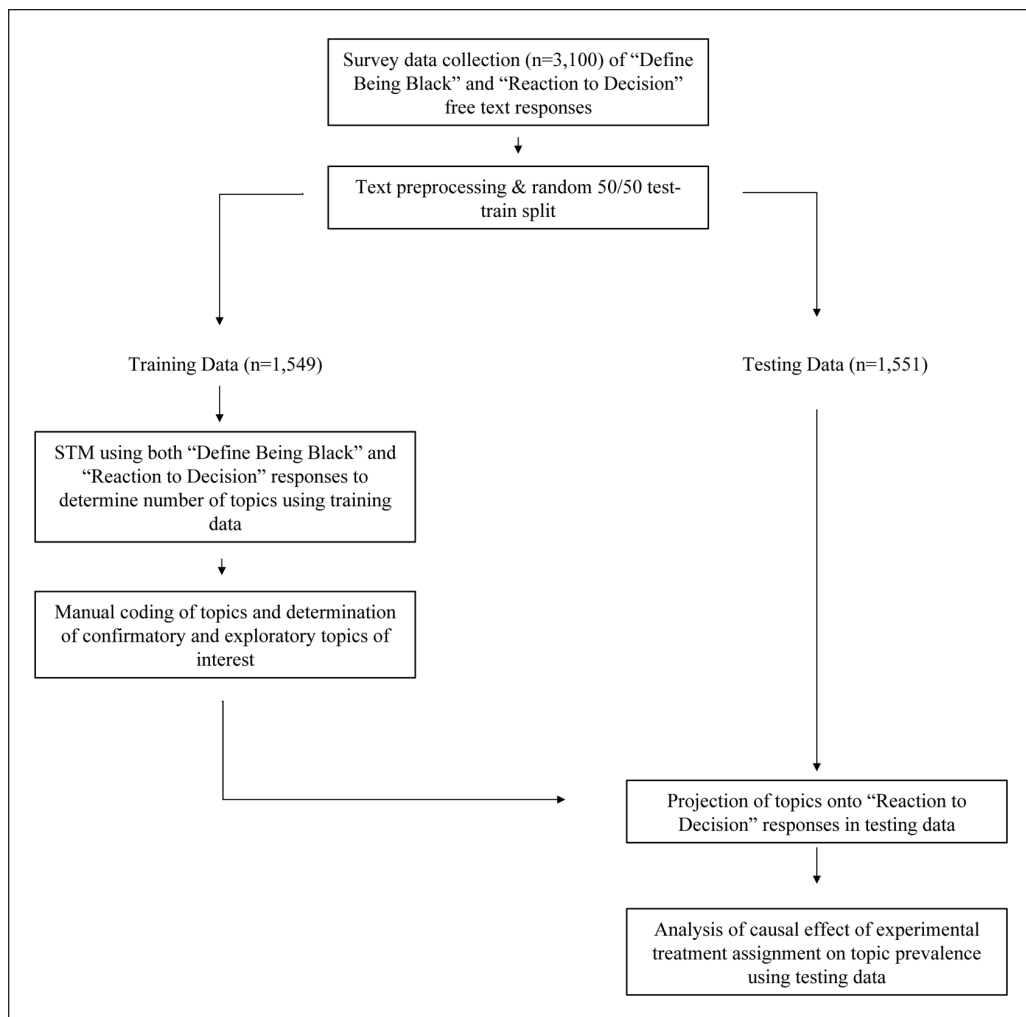


Figure 1. Flowchart of methodological approach to text analyses.

models of different sizes and uses multiple methods to measure the quality of the model with a given number of topics. From there, the metrics are standardized and the model that best fits all four metrics is presented as having the optimal number of topics among the tested parameters. Once this process was complete, we generated the final model which contains the same covariates as our regression on quantitative outcomes variables (described above) and the suggested number of topics.

The training sample contains 1,549 respondents, each of whom provided written answers to two different prompts. We first transformed

the individual data so that each row was now associated with an individual-question ($2 \times 1,549 = 3,098$ documents). Our preprocessing procedure dropped a small number of documents, leaving us with 3,088 to use in our cleaned training sample: 1,546 of these responses were written for the "Define Being Black" prompt, and the other 1,542 were written for the "Reaction to Decision" prompt. Note that while responses to both text questions were combined into a single STM model fit on the training data, the eventual projection of these topics onto the testing set was conducted separately for the "Reaction to

Decision” responses and the “Define Being Black” responses (to allow for a different average prior in each projection). Though we incorporated the “Define Being Black” prompt into the STM model to pull out important themes and topics that may have arisen, we were primarily interested in the effects of our treatments on the “Reaction to Decision” prompt given our theoretical motivation in understanding how respondents evaluated the racial identification of the vignette individual.

Selecting Topics. We determined that an appropriate number of topics for our corpus of data was approximately 45 (see Supplemental Figure A2); this information was then used in our STM model. The model also contained a set of covariates in addition to a binary variable indicating which prompt a document corresponded to (in order to help distinguish effects and topics for each type of response). After generating 45 topics with the training data, we collectively assessed which topics merited further analytical attention. Selecting topics of interest is a primarily qualitative, inductive decision process. We used a multi-step process to isolate our confirmatory and exploratory topics. First, the entire authorship team examined the 10 most prevalent documents for each of the 45 topics to develop a tentative topic name and identify substantively similar topics that might be combined into composite topics. Collectively, we examined each of the 45 topics in turn and corroborated our qualitative designations for each of them. As a check on the similarities between topics under consideration for combination, we also examined the correlation between topics and privileged those with higher degrees of overlap, prioritizing those whose correlation was among the highest 5 percent ($\rho \geq 0.18$) of all pairwise topic correlations.

Based on a combination of substantive interest, coherence, and meaningful relatedness to the research topic, as well as passing the correlation threshold in the case of composite topics, we ended up with 26 qualitatively meaningful topics from the 45 generated from the model. Our final list included five composite topics. Given the multiplicity concerns raised by examining all the 26 topics across GAT results, context, and prior identification

conditions, we chose to designate the topics for either confirmatory or exploratory analyses. To determine which of the 26 to prioritize for confirmatory analyses, we developed a list of our highest priority theoretical concepts (i.e., “Genetic Ancestry”) and those that had meaningful treatment effects in the training data. Of these 26, we prioritized 7 topics for our confirmatory analyses and conducted exploratory analyses using the remaining 19. Note that we selected confirmatory topics based on theoretical relevance and training set effects rather than simply by overall prevalence. Some topics, while prevalent, did not necessarily provide theoretically generative information (e.g., several respondents in the scholarship treatment condition wrote documents that were classified under the Scholarship topic).

We then used equation (1) to estimate the causal effect of each of the high SSA ancestry, scholarship, and Black prior identification conditions on respondent-level prevalence for each topic. We estimated versions of the following equation:

$$y_{ijk} = \beta_0 + \beta_1 \text{Ancestry}_{ijk} + \beta_2 \text{Context}_{ijk} + \beta_3 \text{Prior Identification}_{ijk} + \mathbf{X}_{ijk} \boldsymbol{\Phi} + \pi_j + \gamma_k + \varepsilon_{ijk} \quad (1)$$

where y_{ijk} is the outcome of interest (topic prevalence for a given topic). Ancestry_{ijk} is a treatment indicator for respondent i viewing a vignette with image j and name k for the assigned ancestry condition (either 4 or 36 percent), Context_{ijk} is a treatment indicator for the census versus scholarship conditions, and $\text{Prior Identification}_{ijk}$ is a treatment indicator for the Black versus non-Black prior identification conditions. $\mathbf{X}_{ijk}$ is a vector of individual-level covariates, including demographic characteristics collected pretreatment from the YouGov panel (age, gender, income, ethnicity [Hispanic vs. non-Hispanic], education, political leanings, parent/grandparent immigration history, and state of residence). Finally, π_j is a fixed effect for vignette image and γ_k is a fixed effect for vignette name.

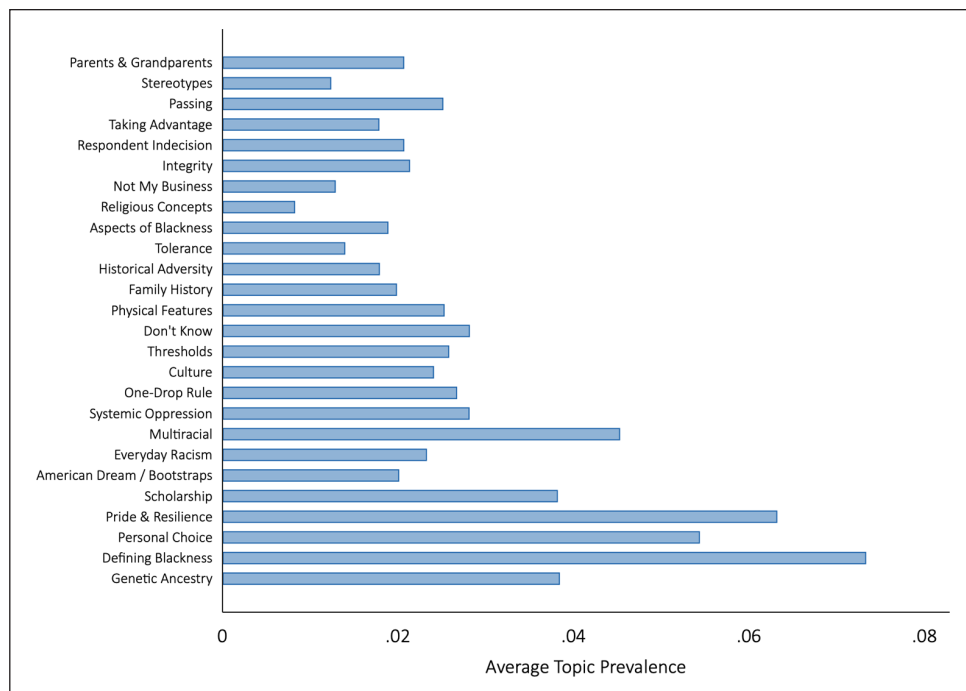


Figure 2. Average prevalence across all documents.

Note. This figure includes responses to both the *Define Being Black* and *Reaction to Decision* prompts across testing and training datasets. Topic prevalences for each document sum to 1.

Results

A full description of all topics, including the most frequent and exclusive (FREX) words and illustrative quotes, can be found in the Supplemental Appendix (Table A3). Figure 2 illustrates the most prevalent topics across the entire data corpus (with a corresponding figure for each prompt separately in the Supplemental Appendix). As can be seen in the figure, the most common topics tended to center a few key themes. Among them are defining the traits socially associated with Blackness (“Defining Blackness”), pride in the Black community (“Pride & Resilience”), discussions of race as a personal choice (“Personal Choice”), and multiracial populations (“Multiracial”).

How do Black Americans define what it means to be Black in America today, and what do these definitions signify for how individuals on the margins ought to identify? We begin with a brief discussion of the themes and narratives that emerged in response to the open-ended prompts, with a focus on those that were

(1) particularly prevalent and (2) prioritized for confirmatory analysis. As can be seen both in Figure 2 and in Supplemental Figure A1, we observe a wide range of perceptions of what it means to be a Black American.

First, the most prevalent topic in the overall data corpus was Defining Blackness. Documents classified under the Defining Blackness topic focused on the role that societal acceptance, upbringing, and collective understanding each play in shaping the definitions of Blackness. For example, one respondent explained that

Being black is having a collective experience with people who are brown and/or black and grow up in brown and/or black households. They are recognized as Black by society. As Black people, they will experience both personal and systemic racism. That the defining factor.⁶

Other respondents had differing definitions rooted in pride or lineage that were also

clustered under the broad Defining Blackness topic. For example, one respondent remarked that “Being black is having a color to you. Black is being proud to be black . . .” while another defined Blackness in simple terms: “Having a black parent.”

Furthermore, in documents primarily classified under the Multiracial topic, respondents frequently mentioned a perception that the vignette individual ought to identify as multiracial rather than as Black. For example, in one such representative document, a respondent wrote,

In actuality, Sam should identify with a mixed race, due to the fact that she identifies as African American but is in actuality mostly of non-African American decent and is 94% another race and 4% African American.

Along similar lines, another remarked that

It’s unfair that if neither of her parents are Black/African American, she shouldn’t identify as a race that her parents don’t identify as . . .

Here, both respondents focus on descent (broadly defined) to make a case that, rather than self-identifying as Black, the vignette individual should identify using a different racial label.

Next, we provide details on the remaining confirmatory topics, in order to provide background for subsequent results regarding the treatment effects of GAT information, context, and prior identification on “Reaction to Decision” responses. As noted in the methods section, we uncovered 26 qualitatively meaningful topics, of which we pre-registered 7 for confirmatory analyses (leaving the remaining 19 as exploratory). The seven confirmatory topics cover a range of different subjects and themes: Historical Adversity, Physical Features, Culture, the “One-Drop” Rule, Pride and Resilience, Genetic Ancestry, and Personal Choice. We provide a few brief examples of each topic below before moving to the experimental treatment effects in the next section. While excerpts below include documents from both

sets of free-response prompts, we focus our treatment effect analysis on only the “Reaction to Decision” prompt.

Respondents whose documents were grouped into the Historical Adversity topic tended to note the Black American community’s long-standing struggle for equality. These documents included language about the history of racism and slavery in the United States, along with contemporary examples that respondents connect to the long U.S. legacy of inequality. For example, when writing about how they define being Black, one respondent noted,

The African American experience is years of slavery, oppression, calculated methods set in place by slave masters to keep Black people oppressed, during, and after slavery, and even to this day.

Likewise, another respondent explained,

Basically what it means to be back [sic] to me is constant struggle day to day with equality or having to look over your shoulder, avoiding the police, being profiled by mostly whites. Constantly struggling with not being able to get a job, pay your bills, get yourself nice things.

To these respondents, the experience of being Black is marked by historical and ongoing oppression and a long history of the struggle for equality in America that continues into the present day.

Pride and Resilience—another topic identified by the STM model—also focused on hardships faced by the Black community. However, descriptions tended to be framed in terms of perseverance and community dignity. For example, one respondent wrote, “To be Black is to be resilient, beautiful, strong, graceful, defiant in the face of a torrid history, current turmoil and an uncertain future.” Similar to the Historical Adversity topic, respondents invoking the topic of Pride and Resilience often referred to the past when making qualifications about what it means to be Black in today’s America. However, this topic tended to be focused more on resilience and overcoming struggle than on adversity itself.

In my opinion, to be Black/African-American in this melting pot commonly known as The United States of America is considered a privilege and honor! We have emerged, as a people, and due to refined liberties and justice our characteristics, experiences, attitudes, and values, combined, enriches the path forward of black excellence . . . Not to sound stereotypical, but I think my culture is more aware of our surroundings and the ongoing resilience it takes, to social climb, during our experiences despite the attitudes of others . . . I'm so proud of how my ancestors paved the way for a better today and more equality tomorrow. We have a bright future!

Cultural experiences also emerged as an important marker of a Black racial identity for respondents. Responses that measured high in the Culture topic focused on the culture in which the vignette individual was presumed to have been raised (i.e., the Black community or the community of another racial group). As an example, one respondent reflected,

If she feels that she was born with the [Black] culture then that is ok. She should celebrate that part of her if it is there and she is raised by that part but also celebrate her other cultures too.

Another explained,

I feel he should go with how he was raised. The way he himself feels about his being. Life can be a feeling of how you carry and feel about who you feel you are.

How respondents invoking the Culture topic tended to view the vignette individual hinged on respondents' assumptions about the vignette individual's cultural upbringing.

Our STM analysis also identified Physical Features, a topic that focused on physical appearance and other features stereotypically associated with individuals racialized as Black, including references to skin tone, hair color, and complexion. Respondents used this topic both to affirm the vignette individual (e.g., "Though she has fair skin, I don't discriminate whos [sic] black or not based on the color of skin") and to reject them (e.g., "She

does not have the phenotypes to be addressed as someone who is black/African-American"), based on their perceptions of the image shown of the vignette individual.

A number of respondents specifically referenced the One-Drop Rule topic in reacting to the vignette individual. For example, in response to the "Define being Black" prompt, one respondent noted, "the one-drop rule that ruled that if a person had one drop of black blood they were black" while another clarified that: "A person being black is a person that have black traits in there [sic] blood. They have black in there [sic] blood." Similar to the Historical Adversity and Pride & Resilience topics, responses in this topic often referenced historical laws when making claims. At the same time, documents classified under the One-Drop Rule topic were distinct in their descriptions of "blood" as a way to discern Blackness.

In addition, many respondents invoked language that was captured by the Genetic Ancestry topic. This topic encapsulated confirmations and rejections of the vignette individual's decision that were based on how respondents appraised the importance or usefulness of the GAT result in shaping one's racial identity. For example, one respondent wrote,

Since she took a genetic ancestry test and her results only showed 4% sub-Saharan ancestry she should not claim to be Black.

Another made clear that, in their opinion:

A % of her DNA shows she's part African and a higher % shows she's non- African, I'm sure the application doesn't care if you're 20/80 or 60/40 somewhere in there is the DNA of an African American, I feel Sam should apply [for the scholarship]!

Finally, several respondents viewed racial identification as a personal choice and something that individuals can define for themselves; this was consolidated into a single topic: Personal Choice. For example, in responding to the vignette individual's decision, one respondent wrote, "He can if he

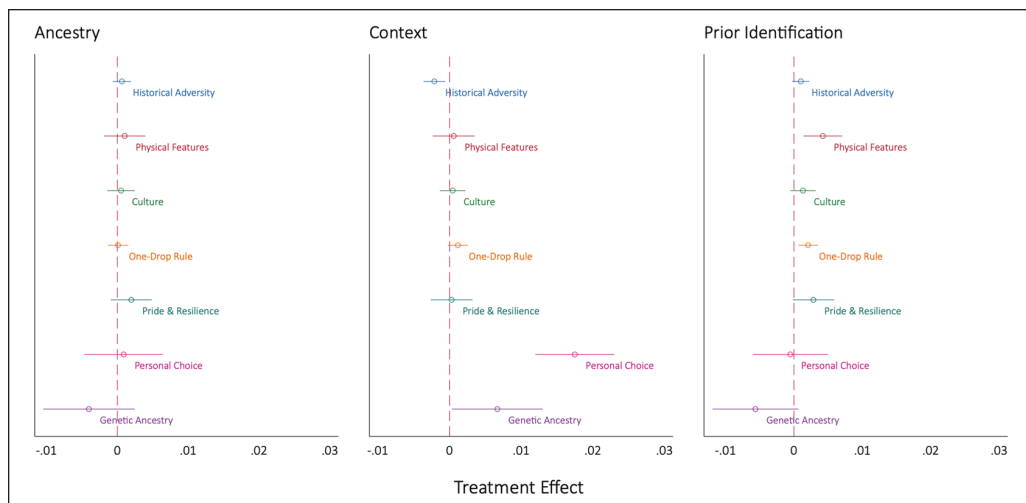


Figure 3. Experimental treatment effects on prevalence of topics in text responses (confirmatory). *Note.* The ancestry panel illustrates the effect of 36 percent SSA relative to 4 percent SSA. The context panel illustrates the effect of the census condition relative to the scholarship condition. The prior identification panel illustrates the effect of a Black prior identity relative to a non-Black prior identity. We estimate treatment effects on topic prevalence for the Reaction to Decision prompt using only the testing set.

wants. Nobody should stop him.” In these documents, respondents tended to deemphasize any characteristics of the vignette individual and instead focus on the personal nature of racial identification decisions.

Effects of GAT Results, Context, and Prior Identification on Topic Prevalence

While the topics identified by the STM model help us to understand the themes and subjects that emerge when respondents are asked to consider GAT results, context, and prior racial self-identification, it is unclear how the various experimental treatments might have affected the topics that respondents used in their evaluations. Figure 3 shows the treatment effect of each of the three experimental conditions (GAT result, context, and prior identification) on the prevalence of a given topic on respondents’ “Reaction to Decision” free responses, which details reactions to the vignette individual’s decision to identify as Black. Tabulated treatment effects can also be found in the Supplemental Appendix (Table A4).

Ancestry information (4 vs. 36 percent SSA) did not shift document topic prevalence

for any of the confirmatory topics of interest, even though it affected classification and Likert-measured approval (Supplemental Table A1). In contrast, we find that respondents in the census condition were more likely to invoke the Personal Choice topic ($\beta = 0.017, p < .001$) and the Genetic Ancestry topic ($\beta = 0.007, p < .05$) in their responses relative to those in the scholarship condition. They also wrote less about the Historical Adversity topic ($\beta = -0.002, p < .01$) in their documents. Moreover, we also observe a statistically significant treatment effect of prior identification: respondents reading a vignette about an individual who previously identified as Black wrote documents classified more prominently under the Physical Features topic ($\beta = 0.004, p < .01$) or the historical “One-Drop” Rule ($\beta = 0.002, p < .01$) than individuals who read a vignette about someone who did not previously identify as Black.

Taken together, these results suggest that the strategies used to draw boundaries around Blackness vary along context and prior identification lines, where respondents used distinct sets of reasonings when confronted with different scenarios. For example, when classification

occurs in the lower-stakes census context, respondents justify their responses through more individualist personal choice rhetoric and refer to the results of the GAT; in contrast, when material resources such as scholarships are at stake, they instead rely on collective histories or oppression to explain their views. In addition, that the vignette individual's stable self-identification as Black activates justifications focused on physical features and the one-drop rule suggests that respondents use different logics based on the perception of racial boundary crossing. Here, assessments of one's phenotype and the logics of hypodescent vary based on whether the vignette individual has had a stable self-identification in the past.

Association between Topic Prevalence, Approval, and Classification

Do the expressed topics have a relationship to (1) approval of the vignette individual, or (2) classifying the vignette individual as Black? In this next section, we turn to two quantitative outcomes measuring reactions to the vignette individual. First, we examine the association between Likert scale approval/disapproval of the vignette individual's decision and written document topics. Second, we examine the association between respondents' classification of the vignette individual as Black and their written topic documents. These results are descriptive and exploratory, intended to broaden understandings of the complex evaluation process that respondents underwent as they determined their reactions to the vignette individual. Furthermore, given the tight correlation between approval and classification outcomes, the results in Figure 4 tend to be similar for both outcomes.

Figure 4 illustrates the association between approval and classification for each of the seven key topics, controlling for ancestry information, context, prior identification, and other covariates. As can be seen in the figure, the only topic that approval and classification are negatively associated with is Genetic Ancestry ($\beta = -0.004, p < .05$ for approval and $\beta = -0.008, p < .05$ for classification, respectively). On the

other hand, approval is positively associated with the Pride and Resilience topic ($\beta = 0.002, p < .05$) and both approval and classification are positively associated with the Personal Choice topic ($\beta = 0.005, p < .001$; $\beta = 0.007, p < .05$). Put simply, these results suggest that those expressing pride in the resilience of the Black community or who express that race is a personal choice may be more likely to approve of the vignette individual or to classify them as Black, net other factors in the model. On the other hand, those who narrate their decision through the lens of genetic ancestry may express lower levels of approval and may be less likely to classify the vignette individual as Black.

Discussion

How do Black Americans construct narratives about Blackness and evaluate who can—and cannot—self-identify as Black in contemporary America? In this study, we move beyond simple quantitative measures of agreement/disagreement or classification to provide a nuanced portrait of the themes and narratives that are deployed in the service of validating or rejecting a person's racial self-identification. All told, our results paint a complex portrait of the ways that Black Americans discuss race and the boundaries of Blackness. We find that discussions tended to focus on a few key concepts, such as race as a personal choice, multiracial populations, and defining the traits that are socially associated with Blackness. However, we also observe wide heterogeneity across these themes, with some respondents instead leaning on discussions of genetic ancestry, the one-drop rule, or even pride and resilience.

In examining the causal effects of treatment assignment on conceptions, several important stories emerge. We do not find a causal effect of the GAT result treatment condition on the seven confirmatory topics. We do, however, find that both the context and the vignette individual's prior identification affected topic usage. Those in the census condition (relative to those in the scholarship condition) were more likely to emphasize race as a personal choice and to discuss genetic ancestry and perceptions of its meaning, and

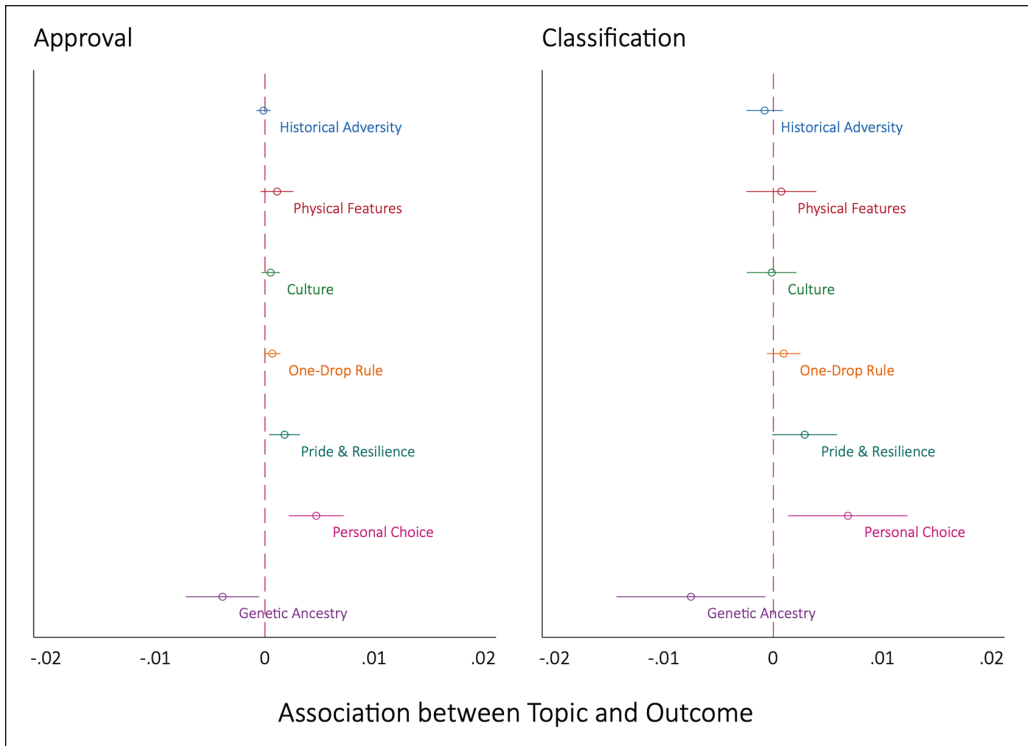


Figure 4. Association between topic prevalence and both approval and classification outcomes.

Note. Approval is measured on a 7-point Likert scale (from “extremely negative” to “extremely positive”); we display standardized values. Classification is the binary outcome of whether the respondent classified the vignette individual as Black. We focus these analyses on the *Reaction to Decision* prompt using only the testing set.

less likely to discuss historical adversities that the Black community has faced. Along similar lines, those reading about a vignette individual who previously identified as Black were more likely to discuss the physical features associated with Blackness and the legacy of the one-drop rule. Finally, several topics were also associated with approval and classification: topics such as race as a personal choice were positively associated with approval and classification as Black, while topics such as genetic ancestry were negatively associated with both approval and classification. In other words, results suggest that those who approve of the vignette individual and who classify the vignette individual as Black may be more likely to frame race as a personal choice. On the other hand, we find that negative views regarding the vignette individual are associated with discussing the situation by relying on the theme of genetic ancestry.

Our results shed light both on the range of justifications used to validate or rebuff another’s racial self-identification, as well as the extent to which such narratives are sensitive to the specifics of the situation. Importantly, we show that beliefs that characterize race as a personal choice or that emphasize one-drop rules are not necessarily inherent, but rather that they are constructed based in part on the context and stakes of the situation. Further, we illustrate that the cues and characteristics that affect classification may not have straightforward effects on the narratives that people use to justify their decision-making to themselves and others. Rather, respondents are selective in which information they integrate into their internal accounts of why a particular individual should (or should not) self-identify as Black. Thus, a complete view of the racial schemas held by contemporary Americans will require attention to both categorization and cultural legitimization.

Future research might continue to disentangle these patterns to examine, for example, the extent to which these differences are strategic on the part of respondents and how other meaningful racial cues (such as skin color or hair texture) affect evaluation and classification processes differently.

In sum, we find that racial boundaries are situationally enacted through shifting justifications, contributing to empirical and theoretical projects in sociology on how such boundaries are formed, negotiated, and permeated (Alba 2005; Gieryn 1983; Okamoto 2014; Wimmer 2008). Indeed, we find that individuals invoke different characteristics—Personal Choice, Pride & Resilience, or Genetic Ancestry—based on the institutional context and perceptions of boundary crossing. As a result, we show that underlying schemas are not fixed but situationally dependent. Further, we also build upon prior calls to develop a sociology of racial appraisals (Roth 2018) which delineates the ways that information from GAT results, paired with other relevant features, shape both reactions by observers as well as their perceptions of the broader social order. These findings support prior work emphasizing that processes of racial identification and classification, at times, require collective input (Brubaker 2016; Roth 2016). Our methodological approach, which combined causal frameworks and quantitative outcomes with recent text-as-data methods, was integral to formulating this insight. Survey experiments that only use one modality, rather than a range of modalities and methods for analysis, may miss important nuances among respondents as they grapple with key social questions (see also Schachter and Weisshaar 2025).

There are several important limitations to this work that merit additional discussion and scrutiny. First, though large-scale survey experiments provide an opportunity to statistically examine responses to different types of fictitious vignette individuals, there may be notable differences between what people report on a survey and how they behave in real-world contexts (Jerolmack and Khan 2014). Much of the information used in our

study is not necessarily information that one would have on hand when classifying a casual passerby; as a result, our findings are but a first step in disentangling how racial classification and evaluation occur. Second, our interpretation of the topic models and their basis may not precisely match respondents' intentions. We aimed to address this by coding all of the topics individually before discussing them (and reaching consensus) between the four authors of this study, though we also acknowledge room for different interpretations of the identified topics. Third, we intentionally limited this study to a small range of images that were similar in their White-Black racial ambiguity. Importantly, while this approach allowed us to estimate how GAT information, context, and prior identification function over a range of similar appearances (all else equal), our results are not generalizable to the wider population of individuals who may claim a Black racial identity or may be racialized as Black by others. Nonetheless, studying how the individuals on the margins of these identities are received can help us to understand the underlying social processes at play. Further research might also consider how a continuous range of SSA ancestry percentages might shape classifications to uncover any tipping points. Finally, while our analyses highlight the racial schemas through which Black Americans conceptualize race and belonging, we acknowledge that we are capturing attitudes toward racial structures rather than the broader structures themselves. Race is a social and political construct that both shapes—and is reproduced through—power relations between groups, social and economic conditions, and institutions, among others (Omi and Winant 2014; Sewell 1992; Vincent 2026). Future research might examine the ways in which social structure and material constraints shape the availability of particular racial categories and meanings in peoples' minds.

Ultimately, our findings reveal that racial boundaries are maintained not through consistent application of fixed rules and explanations, but rather through the flexible deployment of context-specific justifications.

ORCID iDs

Marissa E. Thompson  <https://orcid.org/0000-0002-9497-1400>

A. J. Alvero  <https://orcid.org/0000-0003-4642-9788>

Sam Trejo  <https://orcid.org/0000-0002-9880-5354>

Daphne O. Martschenko  <https://orcid.org/0000-0002-9046-5893>

Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This research was supported in part by seed funding from the Columbia Population Research Center at Columbia University. Results, information, and opinions solely represent the analysis, information, and opinions of the authors and are not endorsed by, or reflect the views or positions of, the grantors.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Supplemental Material

Supplemental material for this article is available online.

Notes

1. Following prior work (Abascal, Armenta, and Halm 2025; Roth 2018; Thompson et al. 2026), we define classification as categorical sorting of people into one race or another (e.g., classification as Black, classification as White). On the other hand, *racial self-identification* describes the race/ethnicity that a given person self-identifies with in categorical terms. For simplicity, we use “racial identification” and “racial self-identification” interchangeably in this study.
2. We fielded this study in early 2023, before the Supreme Court overturned the use of race-based affirmative action in higher education and long before the second Trump Administration’s 2025 rollback of higher education programs supporting Black students.
3. We preregistered this study on the Open Science Framework (<https://osf.io/dj9pt/>). Portions of our preregistered analyses (such as the non-text outcomes) have been published in

a separate, standalone article (see Thompson et al. 2026). In addition, a replication package for the main text analyses can also be found on OSF using the link above.

4. Note that names varied randomly across 12 conditions (6 male and 6 female treatments). We include name and image fixed effects in the models. For clarity and simplicity, we have chosen one of the names to use in the description of the example vignettes.
5. We used birth data for New York state, which includes counts of newborn names by race/ethnicity. Restricting to names with at least 100 counts of Black births and 100 counts of White births, we selected names that were approximately equally likely to be given to a Black newborn compared to a White newborn. More specifically, we opted to include names that had Black/White proportions that matched the overall proportions of Black/White births in New York state. This led us to select the following names to use in the experiments: Samantha (Sam), Mia, Zoe, Isabella, Emily, Ava, Samuel (Sam), Michael, Alexander (Alex), Aaron, Andrew, and Daniel.
6. Note that we present all respondent-provided text verbatim after minimal preprocessing, which includes any misspellings. Text excerpts and their accompanying experimental condition are meant to provide an example of the types of responses included under a given topic. However, they are not meant to be representative of all respondents in a given experimental condition (4 percent vs. 36 percent SSA ancestry/census, vs. scholarship/Black, vs. non-Black prior identification).

References

- Abascal, Maria, Amada Armenta, and W. M. Halm. 2025. “Who Counts as a ‘Person of Color’? The Roles of Ancestry, Phenotype, Self-identification, and Other Factors.” *Socius* 11:1–22. doi:10.1177/23780231251389022.
- Abascal, Maria, Amada Armenta, W. M. Halm, and Daniel J. Hopkins. 2025. “Who Polices Which Boundaries? How Racial Self-identification Affects External Classification.” *American Journal of Sociology* 131(3):630–83. doi:10.1086/737164.
- Abascal, Maria, Amada Armenta, W. M. Halm, Daniel J. Hopkins, and Gall Sigler. 2024. “Calling the Kettle White: How Material Stakes Impact External Racial Classification.” *SocArXiv* 879j_v2. doi:10.31235/osf.io/879j_v2.

- Alba, Richard D. 2005. "Bright vs. Blurred Boundaries: Second-generation Assimilation and Exclusion in France, Germany, and the United States." *Ethnic and Racial Studies* 28(1):20–49. doi:10.1080/0141987042000280003.
- Bailey, Stanley, Fabrício Fialho, and Mara Loveman. 2018. "How States Make Race: New Evidence from Brazil." *Sociological Science* 5:722–51. doi:10.15195/v5.a31.
- Boutyline, Andrei and Laura K. Soter. 2021. "Cultural Schemas: What They Are, How to Find Them, and What to Do Once You've Caught One." *American Sociological Review* 86(4):728–58.
- Brubaker, Rogers. 2016. *Trans: Gender and Race in an Age of Unsettled Identities*. Princeton, NJ: Princeton University Press.
- Condit, Celeste M. 2019. "Laypeople Are Strategic Essentialists, Not Genetic Essentialists." *Hastings Center Report* 49:S27–37. doi:10.1002/hast.1014.
- Davis, F. James. 2001. *Who Is Black? One Nation's Definition*. 10th anniversary ed. University Park, PA: Pennsylvania State University Press.
- DiMaggio, Paul, Manish Nag, and David Blei. 2013. "Exploiting Affinities between Topic Modeling and the Sociological Perspective on Culture: Application to Newspaper Coverage of U.S. Government Arts Funding." *Poetics* 41(6):570–606. doi:10.1016/j.poetic.2013.08.004.
- Egami, Naoki, Christian J. Fong, Justin Grimmer, Margaret E. Roberts, and Brandon M. Stewart. 2022. "How to Make Causal Inferences Using Texts." *Science Advances* 8(42):eabg2652. doi:10.1126/sciadv.abg2652.
- Feldman, M. W. and R. C. Lewontin. 1975. "The Heritability Hang-up." *Science* 190(4220):1163–68.
- Flores, René D., María Vignau Loría, and Regina Martínez Casas. 2023. "Transitory versus Durable Boundary Crossing: What Explains the Indigenous Population Boom in Mexico?" *American Journal of Sociology* 129(1):123–61. doi:10.1086/725337.
- Ghoshal, Raj. 2025. "Racial Appraisals by White, Black, Hispanic, and Multiracial Americans." *Sociology of Race and Ethnicity* 11(4):477–94. doi:10.1177/23326492241254072.
- Gieryn, Thomas F. 1983. "Boundary-work and the Demarcation of Science from Non-science: Strains and Interests in Professional Ideologies of Scientists." *American Sociological Review* 48(6):781. doi:10.2307/2095325.
- Grimmer, Justin, Margaret E. Roberts, and Brandon M. Stewart. 2022. *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton, NJ: Princeton University Press.
- Haslam, Nick, Louis Rothschild, and Donald Ernst. 2000. "Essentialist Beliefs about Social Categories." *British Journal of Social Psychology* 39(1):113–27. doi:10.1348/014466600164363.
- Jerolmack, Colin and Shamus Khan. 2014. "Talk Is Cheap: Ethnography and the Attitudinal Fallacy." *Sociological Methods & Research* 43(2):178–209. doi:10.1177/0049124114523396.
- Lamont, Michèle and Virág Molnár. 2002. "The Study of Boundaries in the Social Sciences." *Annual Review of Sociology* 28(1):167–95. doi:10.1146/annurev.soc.28.110601.141107.
- Lizardo, Omar, Robert Mowry, Brandon Sepulvado, Dustin S. Stoltz, Marshall A. Taylor, Justin Van Ness, and Michael Wood. 2016. "What Are Dual Process Models? Implications for Cultural Analysis in Sociology." *Sociological Theory* 34(4):287–310.
- Loveman, Mara and Jeronimo O. Muniz. 2007. "How Puerto Rico Became White: Boundary Dynamics and Intercensus Racial Reclassification." *American Sociological Review* 72(6):915–39. doi:10.1177/000312240707200604.
- Ma, Debbie S., Joshua Correll, and Bernd Wittenbrink. 2015. "The Chicago Face Database: A Free Stimulus Set of Faces and Norming Data." *Behavior Research Methods* 47(4):1122–35. doi:10.3758/s13428-014-0532-5.
- Ma, Debbie S., Justin Kantner, and Bernd Wittenbrink. 2021. "Chicago Face Database: Multiracial Expansion." *Behavior Research Methods* 53(3):1289–1300. doi:10.3758/s13428-020-01482-5.
- Miles, Andrew, Raphaël Charron-Chénier, and Cyrus Schleifer. 2019. "Measuring Automatic Cognition: Advancing Dual-process Research in Sociology." *American Sociological Review* 84(2):308–33. doi:10.1177/0003122419832497.
- Morning, Ann. 2009. "Toward a Sociology of Racial Conceptualization for the 21st Century." *Social Forces* 87(3):1167–92. doi:10.1353/sof.0.0169.
- Morning, Ann. 2011. *The Nature of Race: How Scientists Think and Teach about Human Difference*. Oakland, CA: University of California Press.
- Murzintcev, Nikita. 2016. "Idatuning: Tuning of the Latent Dirichlet Allocation Models

- Parameters." R package version 0.2.0. (r-project.org).
- Nelson, Alondra. 2016. *The Social Life of DNA: Race, Reparations, and Reconciliation after the Genome*. Boston, MA: Beacon Press.
- Nelson, Laura K. 2020. "Computational Grounded Theory: A Methodological Framework." *Sociological Methods & Research* 49(1):3–42. doi:10.1177/0049124117729703.
- Okamoto, Dina G. 2014. *Redefining Race: Asian American Panethnicity and Shifting Ethnic Boundaries*. New York: Russell Sage Foundation.
- Omi, Michael and Howard Winant. 2014. *Racial Formation in the United States*. 3rd ed. New York: Routledge.
- Orth, Taylor. 2022. *DNA Tests: Many Americans Report Surprises and New Connections*. London, England: YouGov. Retrieved August 1, 2023 (<https://today.yougov.com/topics/society/articles-reports/2022/02/24/dna-tests-many-americans-report-surprises-and-new->).
- Pepin, Joanna R. and William J. Scarborough. 2025. "What Makes a Decision Fair? Relative Earnings, Gender, and Justifications for Couples' Decision-making." *American Journal of Sociology* 130:735618. doi:10.1086/735618.
- Roberts, Dorothy. 2011. *Fatal Invention: How Science, Politics, and Big Business Re-create Race in the Twenty-first Century*. New York: New Press.
- Roberts, Margaret E., Brandon M. Stewart, Dustin Tingley, Christopher Lucas, Jetson Leder-Luis, Shana Kushner Gadarian, Bethany Albertson, and David G. Rand. 2014. "Structural Topic Models for Open-Ended Survey Responses." *American Journal of Political Science* 58(4):1064–82. doi:10.1111/ajps.12103.
- Roth, Wendy D. 2012. *Race Migrations: Latinos and the Cultural Transformation of Race*. Stanford, CA: Stanford University Press.
- Roth, Wendy D. 2016. "The Multiple Dimensions of Race." *Ethnic and Racial Studies* 39 (8):1398–406. doi:10.1080/01419870.2016.1140793.
- Roth, Wendy D. 2018. "Unsettled Identities amid Settled Classifications? Toward a Sociology of Racial Appraisals." *Ethnic and Racial Studies* 41(6):1093–1112. doi:10.1080/01419870.2018.1417616.
- Roth, Wendy D. and Biorn Ivermark. 2018. "Genetic Options: The Impact of Genetic Ancestry Testing on Consumers' Racial and Ethnic Identities." *American Journal of Sociology* 124(1):150–84. doi:10.1086/697487.
- Roth, Wendy D., Elena G. van Stee, and Alejandra Regla-Vargas. 2023. "Conceptualizations of Race: Essentialism and Constructivism." *Annual Review of Sociology* 49(1):39–58. doi:10.1146/annurev-soc-031021-034017.
- Schachter, Ariela, René D. Flores, and Neda Maghbouleh. 2021. "Ancestry, Color, or Culture? How Whites Racially Classify Others in the U.S." *American Journal of Sociology* 126(5):1220–63.
- Schachter, Ariela and Katherine Weisshaar. 2025. "Survey Experiments in Sociology." *Annual Review of Sociology* 51(1):149–69. doi:10.1146/annurev-soc-083024-070113.
- Sewell, William H. 1992. "A Theory of Structure: Duality, Agency, and Transformation." *American Journal of Sociology* 98(1):1–29. doi:10.1086/229967.
- Thompson, Marissa E. 2026. "Paper, Practice, Ancestry, Culture": Racial Frames and Contested Racial/Ethnic Census Categories." *Social Forces* 104(3):959–81. doi:10.1093/sf/soaf075.
- Thompson, Marissa E., Sam Trejo, A. J. Alvero, and Daphne O. Martschenko. Forthcoming, "Being Black is Not Just Some Costume": How Black Americans Narrate Moral Judgments Regarding Racial Boundary-Crossing." *Social Problems*.
- Thompson, Marissa E., Sam Trejo, A. J. Alvero, and Daphne O. Martschenko. 2026. "Policing the Boundaries of Blackness: How Black and White Americans Evaluate Racial Self-identifications." *American Journal of Sociology* 131(5):1147–81. doi:10.1086/740028.
- Vaisey, Stephen. 2009. "Motivation and Justification: A Dual-Process Model of Culture in Action." *American Journal of Sociology* 114(6):1675–715. doi:10.1086/597179.
- Valentino, Lauren. 2021. "Cultural Logics: Toward Theory and Measurement." *Poetics* 88:101574. doi:10.1016/j.poetic.2021.101574.
- Valentino, Lauren and Evangeline Warren. 2024. "Cultural Heterogeneity in Americans' Definitions of Racism, Sexism, and Classism: Results from a Mixed-methods Study." *American Journal of Sociology* 130:733194. doi:10.1086/733194.
- Vincent, Luna. 2026. "Specifying Race: The Colonial Constitution of Race in a Set-theoretic Framework." *Sociological Theory* 44(1):1–24. doi:10.1177/07352751251395706.

- Warikoo, Natasha K. and Janine De Novais. 2015. "Colour-blindness and Diversity: Race Frames and Their Consequences for White Undergraduates at Elite US Universities." *Ethnic and Racial Studies* 38(6):860–76. doi: 10.1080/01419870.2014.964281.
- Wimmer, Andreas. 2008. "The Making and Unmaking of Ethnic Boundaries: A Multilevel Process Theory." *American Journal of Sociology* 113(4):970–1022. doi:10.1086/522803.
- Yudell, Michael, Dorothy Roberts, Rob DeSalle, and Sarah Tishkoff. 2016. "Taking Race out of Human Genetics: Engaging a Century-long Debate about the Role of Race in Science." *Science* 351(6273):564–65. doi:10.1126/science.1257568.
- Zhang, Simone, Janet Xu, and AJ Alvero. 2025. "Generative AI Meets Open-ended Survey Responses: Research Participant Use of AI and Homogenization." *Sociological Methods & Research* 54(3):1197–242. doi:10.1177/00491241251327130.

Author Biographies

Marissa E. Thompson is an assistant professor of sociology at Columbia University. Her work sits at the intersection of stratification, education, race/ethnicity, and social policy. Broadly, she studies the categories and boundaries that organize social life and their consequences for inequality using a

range of quantitative, experimental, and computational methods.

A. J. Alvero is an assistant research professor with the Cornell University Center for Data Science for Enterprise and Society; he holds departmental affiliations with Sociology and Information Science. His research spans sociological inquiry on culture, race and ethnicity, education, science and technology, and evaluation.

Sam Trejo is an assistant professor in the Department of Sociology and the Office of Population Research at Princeton University, who studies how social and biological factors jointly shape human development across the life course. His research spans two areas: integrating genomic data into the social sciences and using field and natural experiments to understand inequality in health and social outcomes. His work has appeared in journals including *PNAS*, *Science Advances*, the *American Journal of Sociology*, and *Demography*.

Daphne O. Martschenko is an assistant professor at the Laurie J. Girand Center for Biomedical Ethics at Stanford. She works at the intersection of biomedical ethics and the social sciences. The overall goal of her research program is to enhance social and ethical responsibility in the conduct and translation of scientific domains that elicit controversy because of disagreement over their potential to lead to harms and benefits.